

Face Detection, Deblurring and Frontalization

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Abstract

CCTV surveillance systems frequently capture faces under challenging conditions such as non-frontal poses, motion blur, occlusions, and poor illumination, limiting their effectiveness for identification and analysis. Existing pipelines primarily focus on detection and recognition, often neglecting the problem of reconstructing high-quality frontal face representations from degraded inputs.

In this work, we propose a quality-aware generative pipeline for extracting high-quality frontal face images from surveillance videos. Faces are detected in each frame using YOLOv8, followed by a multi-factor quality scoring mechanism that evaluates sharpness, brightness, and spatial resolution to select the most informative face instance. The selected face is then enhanced and frontalized using the Qwen Image Edit model.

Experimental results demonstrate clear improvements in visual clarity, sharpness, and pose alignment compared to baseline approaches. Our observations show that generative models are highly sensitive to input quality, and that selecting optimal frames significantly improves output consistency and realism. This highlights the importance of combining quality-aware selection with generative models in real-world surveillance systems.

1. Introduction

CCTV surveillance systems are widely used in public safety, traffic monitoring, and security applications. However, the captured face images are often of low quality due to motion blur, non-frontal viewpoints, occlusions, and poor lighting conditions. These challenges significantly degrade the performance of downstream tasks such as face recognition and identity verification.

While recent advances in computer vision have improved face detection and tracking, the problem of transforming low-quality, non-frontal faces into high-quality frontal representations remains largely underexplored in practical pipelines. Most existing systems either rely on single-frame processing or do not incorporate intelligent selection of the best available data.

In this work, we propose an end-to-end pipeline that combines face detection, quality-based frame selection, and generative enhancement. Instead of processing all frames equally, our method evaluates multiple frames and selects the most informative face instance based on blur, brightness, and spatial resolution before applying generative frontalization.

Novel Contributions:

- A blur-aware frame selection strategy that combines learned blur estimation with brightness and spatial cues to identify optimal frames from CCTV video streams
- A custom blur detection dataset generated using synthetic motion blur, enabling supervised learning of image quality in surveillance conditions
- An end-to-end pipeline integrating face detection, quality-based frame selection, and generative face frontalization
- Empirical analysis demonstrating that input frame quality significantly impacts the performance of generative frontalization models

2. Models and Background

2.1. YOLOv8 for Detection and Quality Estimation

YOLOv8 [8] is a one-stage object detection model that performs bounding box regression and classification in a single forward pass, enabling real-time performance. It adopts an anchor-free design and improved feature aggregation mechanisms for enhanced accuracy and efficiency.

In this work, YOLOv8 is used in two roles:

Face Detection [4]: A YOLOv8n model is trained on a face dataset to detect faces in each frame of the video. The lightweight architecture allows efficient inference while maintaining reliable detection performance.

Blur Estimation [3]: A separate YOLOv8 model is trained to estimate blur levels in face images. The dataset is synthetically generated by applying motion blur to sharp images, enabling supervised learning of blur characteristics. The predicted blur score is used in the frame selection process.

2.2. Qwen Image Editing Model

The Qwen Image Edit model [7] is a diffusion-based generative model designed for instruction-guided image editing tasks. It belongs to the Qwen family of multimodal models that integrate visual and textual understanding.

In this work, the model is used for face frontalization through prompt-based conditioning. Given an input image and a textual instruction, the model generates a frontalized version while preserving identity and visual consistency.

The prompt used is:

”Frontalize this face, make it directly face forward, don’t change the style of image, don’t hallucinate”

The model is accessed via an API [2] interface, avoiding the need for local deployment of large-scale generative models.

2.3. Discussion

The proposed pipeline combines discriminative and generative models in a complementary manner. YOLOv8 ensures reliable detection and quality assessment, while the generative model enhances structural alignment. This combination highlights the importance of input quality in achieving high-quality generative outputs.

3. Proposed Method

3.1. System Overview

The proposed pipeline processes CCTV video to generate high-quality frontal face images. The system consists of sequential stages including frame extraction, face detection, quality assessment, and generative enhancement. Unlike naive approaches that process all frames, our method selectively identifies the most informative frame for further processing.

3.2. Face Detection

Faces are detected in each frame using YOLOv8n, a lightweight object detection model trained on a face dataset. This enables efficient processing of video streams while maintaining reliable detection performance.

3.3. Frame-wise Face Collection

Instead of tracking identities across frames, the system processes each frame independently. For each detected face, corresponding crops are extracted and passed to the quality evaluation stage. This design simplifies the pipeline while still allowing effective selection of high-quality samples.

3.4. Quality Scoring

Each face crop is evaluated using a multi-factor scoring mechanism that considers:

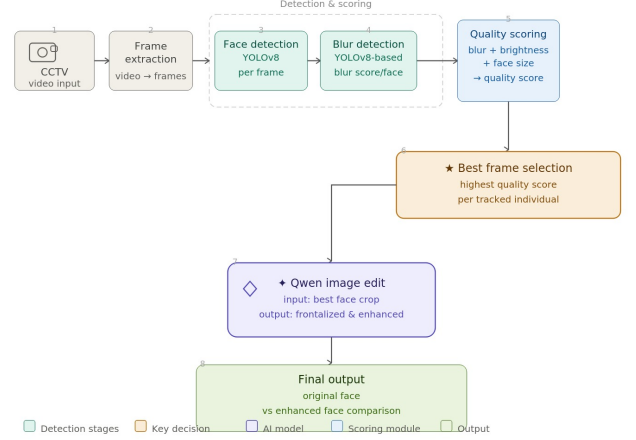


Figure 1. Overview of the proposed pipeline. Faces are detected in each frame, followed by quality-based scoring and selection of the best frame for generative frontalization.

- **Sharpness Score (S_{sharp}):** Estimated using a trained blur detection model, where lower blur corresponds to higher sharpness.
- **Brightness Score (S_{bright}):** Measures illumination conditions to avoid overly dark or saturated frames.
- **Size Score (S_{size}):** Based on the spatial resolution of the detected face region.

The overall score is computed as:

$$Score = w_1 \cdot S_{sharp} + w_2 \cdot S_{bright} + w_3 \cdot S_{size} \quad (1)$$

The frame with the highest score is selected as the optimal candidate for frontalization.

3.5. Generative Enhancement

The selected face image is processed using the Qwen Image Edit model via an API-based interface. A prompt-driven approach is used to guide the model towards generating a frontal face while preserving identity and visual consistency. This step transforms non-frontal, low-quality inputs into structured frontal representations.

3.6. Key Insight

A key insight of this work is that generative models are highly sensitive to input quality. Direct application of frontalization on arbitrary frames often leads to artifacts and inconsistent outputs. By introducing a quality-aware selection stage, the proposed pipeline ensures that only high-quality inputs are processed, resulting in significantly improved realism and structural fidelity in the generated images.

4. Experiments and Results

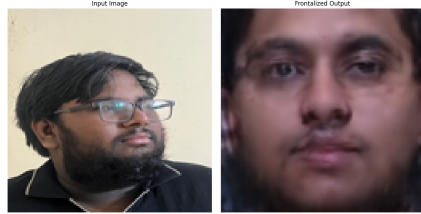
4.1. Experimental Setup

Prior to the final architecture, we experimented with two baseline approaches:

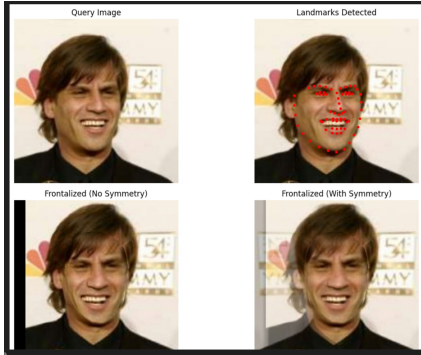
1. **Generative Frontalization (Pre-trained) [6]:** A GAN-based approach using the Hugging Face model, with the architecture defined in `network.py` and weights from `generator.pt`.
2. **Initial Baseline Model [1]:** A custom implementation based on geometric transformations, as shown in the repository.

The **final proposed model** builds upon these approaches by incorporating a multi-stage pipeline including face detection, quality-based frame selection, enhancement, and frontalization.

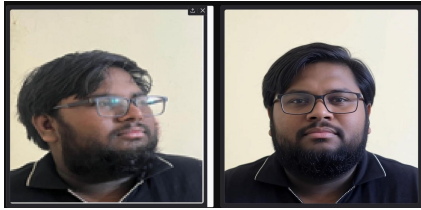
4.2. Qualitative Results



(a) GAN-based Model



(b) Baseline Model



(c) Proposed Model

[5]

Figure 2. Qualitative comparison between baseline approaches and the proposed pipeline.

The qualitative results demonstrate that the proposed method significantly improves facial clarity, sharpness, and

pose alignment compared to both baseline approaches. The final model produces more consistent and visually coherent outputs, particularly under challenging conditions such as motion blur and non-frontal poses.

The proposed model consistently outperforms both the GAN-based and custom baseline models across all evaluated conditions, with the most noticeable improvements observed in moderate blur and pose variation scenarios.

4.3. Observations

- The multi-stage pipeline significantly improves robustness over single-stage models.
- Quality-based frame selection plays a critical role in enhancing final output quality.
- The proposed model performs particularly well under moderate distortions compared to baseline methods.
- Larger and clearer face regions result in better reconstruction performance.
- Failure cases include extreme occlusion and severely degraded low-resolution inputs.

Method	Robustness	Visual Quality
GAN Model	Low	Medium
Baseline	Low	Low
Proposed	High	High

5. Implementation Details

5.1. Development Environment

All experiments were conducted on Kaggle notebooks with GPU acceleration. The training and inference pipelines were implemented using PyTorch and the Ultralytics YOLOv8 framework.

5.2. Hardware Configuration

- GPU: 2 × NVIDIA T4 GPUs
- RAM: 16 GB
- Framework: PyTorch

5.3. Model Configurations

Face Detection:

- Model: YOLOv8n
- Training: Fine-tuned on a face dataset from Kaggle

Blur Detection:

- Model: YOLOv8 (custom trained)
- Dataset: Synthetic dataset generated by applying motion blur to sharp images

Frontalization Model:

- Model: Qwen Image Edit
- Access: Hugging Face API
- Motivation: Local deployment was computationally expensive, making API-based inference more practical

5.4. Inference Pipeline

The pipeline processes video frames sequentially:

1. Face detection on each frame
2. Quality scoring based on blur, brightness, and size
3. Selection of the highest-scoring frame
4. API-based frontalization using Qwen

6. Conclusion

We presented an end-to-end pipeline for face enhancement and frontalization tailored to CCTV surveillance scenarios. The proposed approach integrates face detection, quality-aware frame selection, and generative image editing to produce high-quality frontal face images from challenging inputs.

A key insight from this work is that the performance of generative models is highly dependent on input quality. By introducing a blur-aware scoring mechanism that incorporates sharpness, brightness, and spatial resolution, the pipeline effectively selects optimal frames, leading to significantly improved perceptual quality and structural consistency in the generated outputs compared to baseline approaches.

Unlike traditional methods that directly apply frontalization, the proposed system emphasizes input selection as a critical preprocessing step, making it more robust under real-world conditions such as motion blur and non-frontal poses.

Future work includes incorporating temporal tracking for identity consistency, multi-frame fusion to improve reconstruction quality, and real-time deployment for practical surveillance applications.

7. Limitations

Despite the improvements, several challenges remain:

Identity Drift: Generative models may produce outputs that deviate from the true identity, especially under extreme pose variations or poor input quality.

Texture Smoothing: Reconstructed regions may lack fine-grained details such as skin texture, leading to slightly over-smoothed or synthetic appearances.

Handling Occlusions: Occlusions such as glasses, masks, or hair can introduce artifacts or inconsistencies in the generated outputs.

Computational Overhead: While API-based inference reduces local computation requirements, latency remains a constraint for real-time deployment.

8. Future Work

While the proposed pipeline demonstrates strong performance, several extensions can further improve its effectiveness:

Incorporating Tracking: Integrating a tracking mechanism can enable consistent identity association across frames, allowing selection of the best frame per individual and improving robustness in multi-person scenarios.

Improved Detection Models: Training detection models on CCTV-specific datasets can improve performance under challenging conditions such as low resolution, motion blur, and varying illumination, leading to more reliable face localization.

Model Optimization and Distillation: Distilling the generative model into a smaller and more efficient architecture can enable local deployment and reduce reliance on API-based inference, making the system suitable for real-time and edge applications.

A. Code and Resources

The implementation of the proposed pipeline and trained models are available at the following links:

- Project Notebook: <https://www.kaggle.com/code/safebet1034/ee655-face-frontalization>
- Blur Detection Model: https://www.kaggle.com/models/safebet1034/motion_blur_detection_yolov8/PyTorch/default/1
- Blur Detection Dataset: <https://www.kaggle.com/datasets/safebet1034/blur-deblur>
- Face Detection Model: <https://www.kaggle.com/models/safebet1034/face-detection/PyTorch/default/1>
- Face Detection Dataset: <https://www.kaggle.com/datasets/fareselmenshawii/face-detection-dataset>
- GitHub Repository: https://github.com/Ahmed2062/EE655_Face_Frontalization

B. Author Contributions

The contributions of each team member towards the project are summarized below:

- **Ayush Yadav and Shreyansh Singh:** Conducted literature review and explored existing methods for face frontalization and enhancement, including identifying and evaluating suitable baseline and generative models.
- **Snehasish Halder:** Performed experimental evaluations of different models and contributed to report structuring, formatting, and documentation.
- **Ahmed Adnan:** Implemented and trained the YOLOv8-based face detection module for extracting facial regions from video frames.
- **Suyash Kumar:** Developed the YOLOv8-based blur detection model, integrated the Qwen Image Edit model for frontalization, and designed the overall pipeline including frame selection and system integration.

References

- [1] Dougsouza. Face-frontalization. <https://github.com/dougsouza/face-frontalization>, 2015.
- [2] Hugging Face. Hugging face platform. <https://huggingface.co/>, 2024.
- [3] Suyash Kumar. Motion blur detection using yolov8. https://www.kaggle.com/models/safebet1034/motion_blur_detection_yolov8/PyTorch/default/1, 2026.
- [4] Suyash Kumar. Yolov8 face detection model. <https://www.kaggle.com/models/safebet1034/face-detection/PyTorch/default/1>, 2026.
- [5] Pruthviraj Meghe. Real example, 2026.
- [6] Opetrova. Face frontalization model. <https://huggingface.co/opetrova/face-frontalization>, 2023.
- [7] Qwen Team. Qwen image edit 2511. <https://huggingface.co/Qwen/Qwen-Image-Edit-2511>, 2024.
- [8] Ultralytics. Yolov8 documentation. <https://docs.ultralytics.com/models/yolov8/>, 2023.